555 LED Blinker

Circuit simulation datasheet

Doc No.	CJS-0001
Revision	A
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Author	Your name

1 Description

555 astable LED blinker (about 1 Hz)

2 Schematic

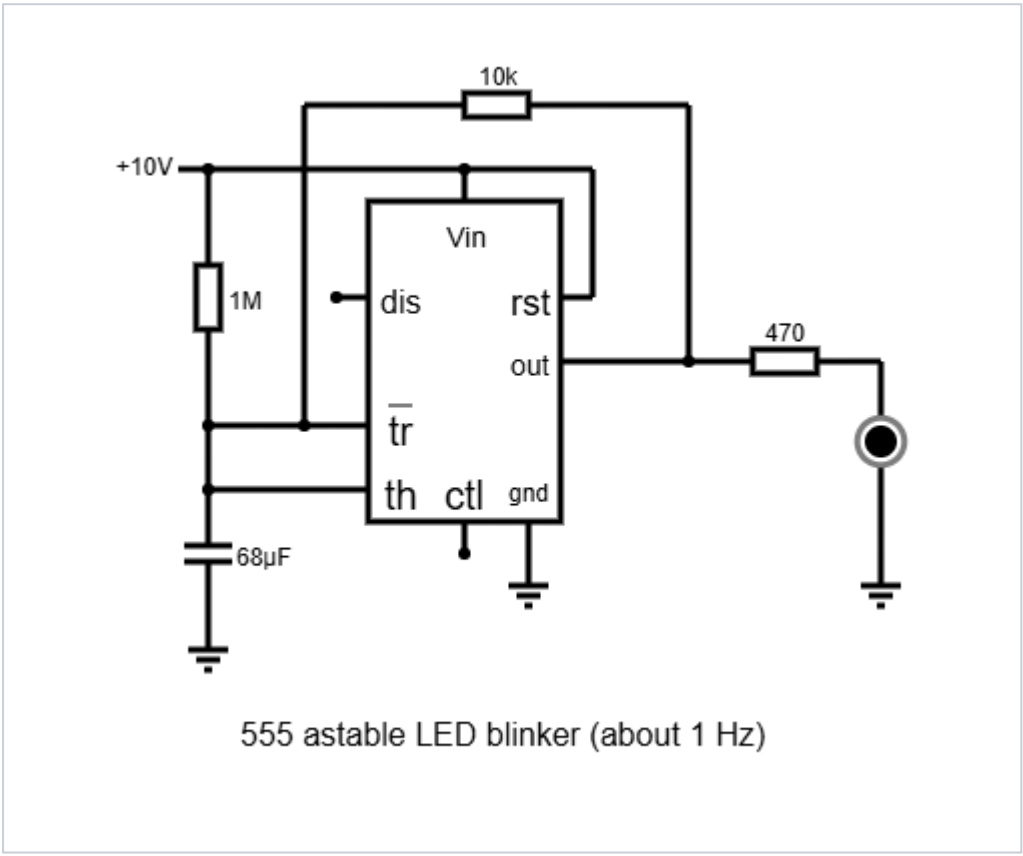


Figure 1. Circuit schematic

3 Electrical Characteristics

Measured over t = 100µs to 3s (the range shown in the Waveform Viewer), simulation time step 100µs. Typ = time-weighted average.

Parameter	Symbol	Min	Typ	Max	Unit
V1: wire					
Voltage	V_1	0.000	5.588	9.982	V
Peak-to-peak	V_{1pp}		9.982		V
RMS	V_{1rms}		7.468		V
Frequency	f		923.9		mHz
Period	T		1.082		s
Duty cycle	D		55.9		%
V2: capacitor, 68 µF					
Voltage	V_2	0.001	4.754	6.667	V
Peak-to-peak	V_{2pp}		6.666		V
RMS	V_{2rms}		4.940		V

4 Typical Waveforms

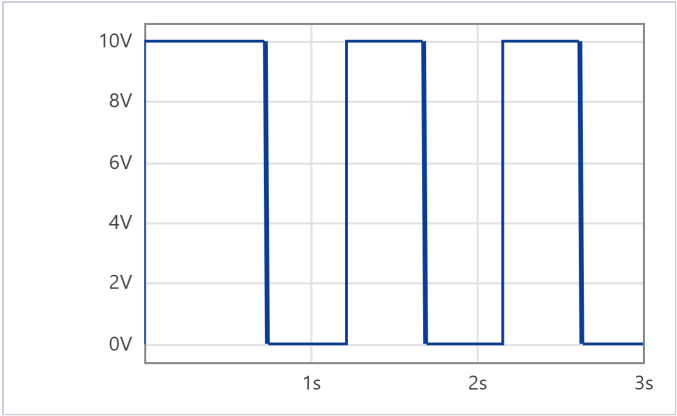


Figure 2. V1: wire vs time

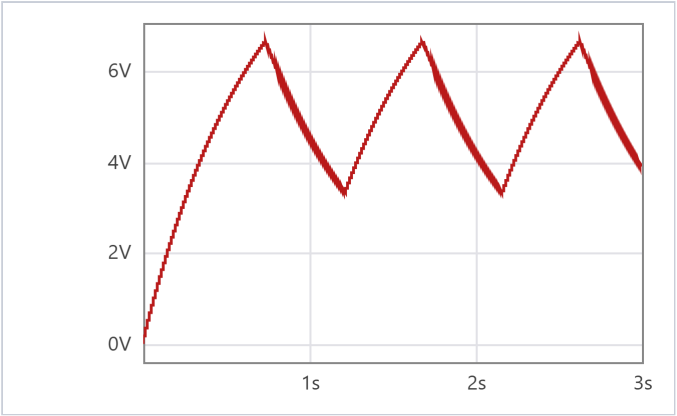


Figure 3. V2: capacitor, 68 µF vs time

5 Bill of Materials

Item	Part	Value	Qty
1	555 Timer		1
2	Capacitor	68 µF	1
3	LED (old-default-led)		1
4	Resistor	1 MΩ	1
5	Resistor	10 kΩ	1
6	Resistor	470 Ω	1
7	Voltage source	10V	1

6 Simulation Conditions

Simulation time step	100µs
Simulated time	3s
Circuit elements	21
Recorded channels	2 (60,000 samples)
Simulator	CircuitJS Desktop (modified nodal analysis, trapezoidal / backward Euler integration)
Generated	10/11/2026, 9:01:39 pm

Values are simulation results from CircuitJS Desktop (ideal or simplified part models, no tolerances or temperature effects). Verify on real hardware before relying on them.

Appendix A Circuit File

Paste into File → Import From Text, or save as a .csim file, to reproduce this simulation.

```
$ 1 0.0001 56.303023683595114 50 5 50 5e-11
w 112 192 112 224 0
w 112 224 176 224 0
c 112 224 112 288 0 0.000068 3.9104276586195335 0.001
g 112 288 112 304 0 0
r 112 192 112 64 0 1000000
w 112 64 240 64 0
R 112 64 80 64 0 0 40 10 0 0 0.5
165 176 96 192 96 6 0.00039100366549540375
w 240 64 304 64 0
w 304 64 304 128 0
w 352 160 304 160 0
w 352 160 352 32 0
r 352 32 160 32 0 10000
w 160 32 160 192 0
w 112 192 160 192 0
w 160 192 176 192 0
g 272 256 272 272 0 0
r 352 160 448 160 0 470
162 448 160 448 240 2 old-default-led 1 0 0 0.01
g 448 240 448 272 0 0
x 128 352 390 355 4 16 555\sastable\sLED\sblinker\s(about\s1\sHz)
```